Category C · Coding Assignment

Problem 7: Second-Best Score — Solution

TASK

A quiz app stores the scores of all players in an array. Prizes go to the top two different scores, so the app needs the second-highest distinct score. If every player has the same score, there is no second-highest score.

Implement a function secondHighest(scores) that returns the second-highest distinct score, or -1 if it does not exist.

CONSTRAINTS

• 1 <= length of scores <= 1000
• 0 <= scores[i] <= 100
• Do not sort the array

IMPLEMENTATION (JavaScript / Python)

```
function secondHighest(scores) {
    let highest = -1;
    let second = -1;
    for (let score of scores) {
        if (score > highest) {
            second = highest;
            highest = score;
        } else if (score > second && score < highest) {
            second = score;
        }
    }
    return second;
}
```

COMPLEXITY REQUIREMENTS & COMPARISON

• **Time Complexity: O(n)**, where *n* is the number of elements in the scores array. We iterate through the array exactly once.
• **Space Complexity: O(1)** auxiliary space, as we only use two variables (highest and second) regardless of the input array size.
• **Comparison with Sorting:** Sorting the array first typically takes **O(n log n)** time and requires additional space depending on the sorting algorithm. The single-pass approach avoids sorting entirely, satisfying the constraint while operating in optimal linear time and constant space.

EXAMPLES VERIFICATION

• **Sample 1:** scores = [45, 78, 92, 78, 60] → Output: **78**
• **Sample 2:** scores = [50, 50, 50] → Output: **-1**